Bittu Kumar

MLOps & LLMOps Engineer

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LinkedIn | GitHub | Blog | Portfolio

Professional Summary

MLOps & LLMOps Engineer with 2 years of experience in building and deploying scalable ML and LLM systems. Experienced in Kubernetes, Docker, AWS, and end-to-end ML lifecycle management, with hands-on expertise in RAG-based LLM pipelines, model serving, and production monitoring.

Technical Skills

LLMOps: RAG, LangSmith, NLP, Pinecone (Vector DB)
MLOps: MLflow, Kubeflow, DVC, Airflow
Model Serving: KServe, FastAPI, Flask, Gunicorn
DevOps & CI/CD: Docker, Kubernetes, Helm, Jenkins, GitHub Actions, ArgoCD
Cloud: AWS (EKS, EC2, S3, IAM, SageMaker)
Infrastructure: Terraform, Ansible
Monitoring: Prometheus, Grafana, CloudWatch
Languages: Python, Bash

Professional Experience

EXL Service India Pvt. Ltd. – Noida

Associate Analyst – MLOps
April 2024 – Present

- Designed and implemented an end-to-end MLOps workflow covering model training, versioning, deployment, serving, scaling, and lifecycle management
- Built and productionized ML models with clear separation between model logic, serving layer, and infrastructure
- Implemented custom model serving using Flask with Gunicorn (WSGI) to understand low-level serving concepts
- Containerized ML workloads using Docker to ensure reproducible builds and immutable artifacts
- Deployed ML models on Kubernetes using Deployments, Services, and Ingress for scalable inference
- Implemented KServe-based model serving using standardized inference APIs
- Evaluated Flask-based serving versus KServe-managed serving, analyzing scalability and operational trade-offs
- Integrated external model artifact storage using AWS S3 and HTTP-based endpoints
- Worked with SageMaker Domain, User Profiles, and IAM execution roles for secure ML environments
- Designed IAM-based role separation for Data Scientists, ML Engineers, and MLOps Engineers
- Applied MLOps automation principles to automate model training, artifact generation, and deployment readiness

- Implemented end-to-end pipeline orchestration using Kubeflow
- Gained hands-on experience with production MLOps responsibilities including reliability, deployment safety, and scalability

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EXL Service India Pvt. Ltd. – Noida

Associate Analyst – LLMOps
April 2024 – Present

- Designed and implemented a production-grade RAG-based LLM pipeline for scalable AI applications
- Integrated vector database using Pinecone for semantic search and context-aware retrieval
- Developed LLM APIs using Python and FastAPI for real-time inference and integration
- Implemented LLM observability using LangSmith to monitor prompt performance, latency, and outputs
- Built evaluation workflows and applied Pytest-based testing to ensure reliability of LLM pipelines
- Containerized applications using Docker and deployed on ECS Fargate and EKS for scalable infrastructure
- Designed and automated CI/CD pipelines using Jenkins and GitHub Actions for continuous integration and deployment
- Implemented GitOps-based deployment using ArgoCD for Kubernetes environments
- Configured monitoring and alerting using Prometheus and Grafana for system performance and observability

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Education

Bachelor of Computer Applications (B.C.A – Computer Science)
Arcade Business College
May 2020 – July 2023

Certifications

- Certified DevOps Professional – Oracle Cloud Infrastructure
- MLOps Training & Certification – EXL Service India Pvt. Ltd.
- LFS147: Introduction to AI/ML Toolkits with Kubeflow – The Linux Foundation

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Technical Writing & Blogging

- Write technical blogs on DevOps, Cloud, MLOps, and AI Engineering
- Publish hands-on articles on CI/CD pipelines, Kubernetes, AWS services, and MLOps workflows
- Focus on practical implementations, system architecture explanations, and best practices